计算机网络实验报告

课程名称 计算机网络 成绩评定

实验项目名称 Wireshark 综合实验（3） 指导教师 张伟

实验项目编号 实验 6 实验项目类型 实验地点

学生姓名 李汶峰 学号 202300460158

学院 网络空间安全 专业 网络空间安全

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一、 实验目的

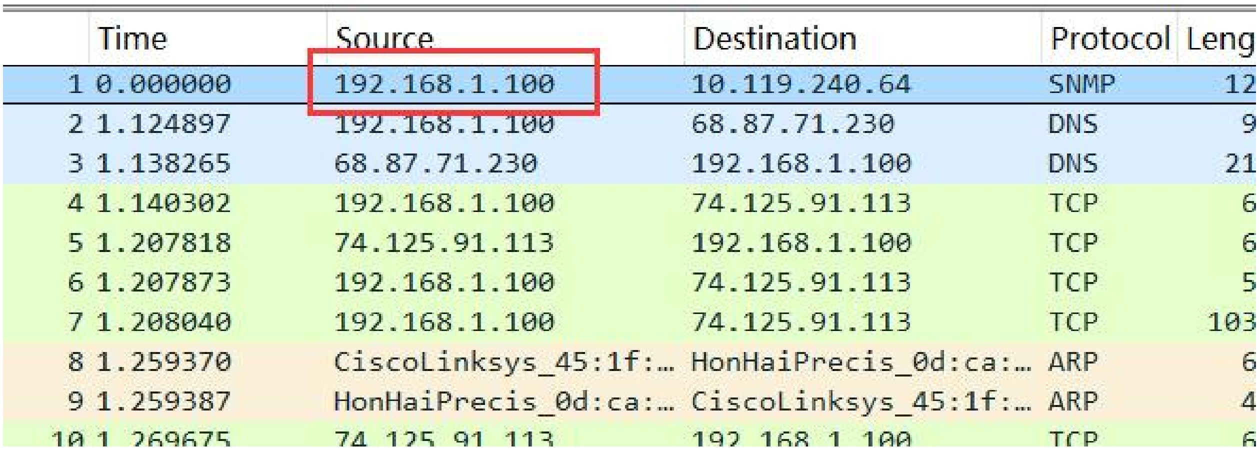
理解 NAT 协议的运作。

二、 实验步骤与结果

任务一：打开 NAT_home_side 文件并回答以下问题

1.客户端的 IP 地址是多少？

如下图所示，客户端 IP 地址为 **192.168.1.100**



	Time	Source	Destination	Protocol	Leng
1	0.000000	192.168.1.100	10.119.240.64	SNMP	12
2	1.124897	192.168.1.100	68.87.71.230	DNS	9
3	1.138265	68.87.71.230	192.168.1.100	DNS	21
4	1.140302	192.168.1.100	74.125.91.113	TCP	6
5	1.207818	74.125.91.113	192.168.1.100	TCP	6
6	1.207873	192.168.1.100	74.125.91.113	TCP	5
7	1.208040	192.168.1.100	74.125.91.113	TCP	103
8	1.259370	CiscoLinksys_45:1f:…	HonHaiPrecis_0d:ca:…	ARP	6
9	1.259387	HonHaiPrecis_0d:ca:…	CiscoLinksys_45:1f:…	ARP	4
10	1.269675	74.125.91.113	192.168.1.100	TCP	6

2.为了仅显示客户端的请求和服务器的响应，请在 Wireshark 过滤器输入以下过滤条“http && ip.addr == 64.233.169.104 ”

截图如下:

http && ip.addr == 64.233.169.104

No.	Time	Source	Destination	Protocol	Lengtl	Info
56	7.109267	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
60	7.158797	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
62	7.281399	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
73	7.349451	64.233.169.104	192.168.1.100	HTTP	226	HTTP/1.1 200 OK (GIF89a)
75	7.370185	192.168.1.100	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMAo4NUAILCswDjgHLCswFjgQLCswFzgDLCswGDgELCswGTgJLCswHTgZLCswJTjJiAEsKzAmOAUsKzAnOAIsKzAqOA
92	7.448649	64.233.169.104	192.168.1.100	HTTP	648	HTTP/1.1 200 OK (text/javascript)
94	7.492324	192.168.1.100	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
100	7.537353	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
107	7.652836	192.168.1.100	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
112	7.682361	192.168.1.100	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17259,21588,21766,21920&ei=25O2Ssb1G4_CeJvxxaMO&rt=prt.128,xjs.287,ol.437
119	7.685786	64.233.169.104	192.168.1.100	HTTP	1359	HTTP/1.1 200 OK (PNG)
122	7.709490	192.168.1.100	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
124	7.737783	64.233.169.104	192.168.1.100	HTTP	269	HTTP/1.1 204 No Content
127	7.763501	64.233.169.104	192.168.1.100	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

3.请选择在 7.109267s 时间的客户端发送到 Google 服务器的 HTTP GET。承载此 HTTP GET 的 IP 数据报上的源 IP 地址和目标 IP 地址以及 TCP 源端口和目标端口是什么?

截图如下：源 IP 地址为 192.168.1.100，目的 IP 地址为 64.233.169.104

TCP 源端口为 4335，目标端口为 80

```
Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 675
    Identification: 0xa2ac (41644)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0xa94a [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.100
    Destination Address: 64.233.169.104
    [Stream index: 5]
Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 4164040421, Ack: 3914283157, Len: 635
    Source Port: 4335
    Destination Port: 80
    [Stream index: 2]
    [Stream Packet Number: 4]
  > [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 635]
    Sequence Number: 4164040421
    [Next Sequence Number: 4164041056]
    Acknowledgment Number: 3914283157
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
```

4.对于前一问发送的 HTTP GET，在什么时间客户端从 Google 服务器收到对应的状态码 200、状态 OK 的 HTTP 响应消息？携带状态码 200、状态 OK 的 HTTP 响应消息的 IP 数据报上的源和目标 IP 地址以及 TCP 源和目标端口是什么?

如下图所示，客户端在第 7.158797s 收到 Google 的 HTTP 相应消息，该消息状态码为 200、状态为 OK.

	Time	Source	Destination	Protocol	Lengtl	Info
56	7.109267	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
60	7.158797	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
62	7.281399	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
73	7.349451	64.233.169.104	192.168.1.100	HTTP	226	HTTP/1.1 200 OK (GIF89a)
75	7.370185	192.168.1.100	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMAo4NUAILCswDjgHLCs
92	7.448649	64.233.169.104	192.168.1.100	HTTP	648	HTTP/1.1 200 OK (text/javascript)
94	7.492324	192.168.1.100	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
100	7.537353	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
107	7.652836	192.168.1.100	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
112	7.682361	192.168.1.100	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17
119	7.685786	64.233.169.104	192.168.1.100	HTTP	1359	HTTP/1.1 200 OK (PNG)
122	7.709490	192.168.1.100	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
124	7.737783	64.233.169.104	192.168.1.100	HTTP	269	HTTP/1.1 204 No Content
127	7.763501	64.233.169.104	192.168.1.100	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

源 IP 地址为 64.233.169.104，目标 IP 地址为 192.168.1.100

TCP 源端口为 80，目标端口为 4335。

```
Internet Protocol Version 4, Src: 64.233.169.104, Dst: 192.168.1.100
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x20 (DSCP: CS1, ECN: Not-ECT)
    Total Length: 800
    Identification: 0xf61e (63006)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 50
    Protocol: TCP (6)
    Header Checksum: 0xe33b [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 192.168.1.100
    [Stream index: 5]
Transmission Control Protocol, Src Port: 80, Dst Port: 4335, Seq: 3914286017, Ack: 4164041056, Len: 760
    Source Port: 80
    Destination Port: 4335
    [Stream index: 2]
    [Stream Packet Number: 8]
  > [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 760]
    Sequence Number: 3914286017
    [Next Sequence Number: 3914286777]
    Acknowledgment Number: 4164041056
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
    Window: 110
```

5.在什么时间客户端发送了含有 TCP SYN 的报文建立连接消息，以后续用于发送在 7.109267 s 的 GET 请求？TCP SYN 报文的源 IP 地址和目标 IP 地址以及源端口和目标端口是什么？在什么时间客户端收到了对应的 SYN-ACK 报文？此 SYN-ACK 报文的源和目标 IP 地址以及源端口和目标端口是什么？

如下图所示，红框划出部分即为 7.109276s 的 GET 请求前的三次握手过程。其中，客户端发送含有 TCP SYN 的报文建立连接消息时间为 **7.075657s**.

该 TCP SYN 报文的源 IP 地址为 **192.168.1.100**, 目标 IP 地址为 **64.233.169.104**

TCP 源端口为 **4335**，目标端口为 **80**

41	1.976996	192.168.1.100	74.125.106.31	HTTP	772 GET /safebrowsing/rd/goog-malware-shavar_a_14466-14470.14466.14467-14470: HTTP/1.1
42	2.000629	74.125.106.31	192.168.1.100	HTTP	881 HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
43	2.014105	192.168.1.100	74.125.106.31	HTTP	776 GET /safebrowsing/rd/goog-phish-shavar_s_48291-48300.48291-48295.48296-48300: HTTP/1.1
44	2.038247	74.125.106.31	192.168.1.100	HTTP	526 HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
45	2.044751	192.168.1.100	74.125.106.31	HTTP	776 GET /safebrowsing/rd/goog-phish-shavar_a_67721-67760.67721-67729.67730-67760: HTTP/1.1
46	2.064877	74.125.106.31	192.168.1.100	HTTP	1089 HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
47	2.178596	192.168.1.100	74.125.106.31	TCP	54 4331 → 80 [ACK] Seq=1474855900 Ack=3015694974 Win=260176 Len=0
53	7.075657	192.168.1.100	64.233.169.104	TCP	66 4335 → 80 [SYN] Seq=4164040420 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
54	7.108986	64.233.169.104	192.168.1.100	TCP	66 80 → 4335 [SYN, ACK] Seq=3914283156 Ack=4164040421 Win=5720 Len=0 MSS=1430 SACK_PERM WS=64
55	7.109053	192.168.1.100	64.233.169.104	TCP	54 4335 → 80 [ACK] Seq=4164040421 Ack=3914283157 Win=260176 Len=0
56	7.109267	192.168.1.100	64.233.169.104	HTTP	689 GET / HTTP/1.1
57	7.140728	64.233.169.104	192.168.1.100	TCP	60 80 → 4335 [ACK] Seq=3914283157 Ack=4164041056 Win=7040 Len=0
58	7.158432	64.233.169.104	192.168.1.100	TCP	1484 80 → 4335 [ACK] Seq=3914283157 Ack=4164041056 Win=7040 Len=1430 [TCP PDU reassembled in 60]
59	7.158761	64.233.169.104	192.168.1.100	TCP	1484 80 → 4335 [ACK] Seq=3914284587 Ack=4164041056 Win=7040 Len=1430 [TCP PDU reassembled in 60]
60	7.158797	64.233.169.104	192.168.1.100	HTTP	814 HTTP/1.1 200 OK (text/html)
61	7.158844	192.168.1.100	64.233.169.104	TCP	54 4335 → 80 [ACK] Seq=4164041056 Ack=3914286777 Win=260176 Len=0
62	7.281399	192.168.1.100	64.233.169.104	HTTP	719 GET /intl/en_ALL/images/logo.gif HTTP/1.1
63	7.315019	64.233.169.104	192.168.1.100	TCP	309 80 → 4335 [PSH, ACK] Seq=3914286777 Ack=4164041721 Win=8320 Len=255 [TCP PDU reassembled in
64	7.315576	64.233.169.104	192.168.1.100	TCP	1484 80 → 4335 [ACK] Seq=3914287032 Ack=4164041721 Win=8320 Len=1430 [TCP PDU reassembled in 73]
65	7.315641	192.168.1.100	64.233.169.104	TCP	54 4335 → 80 [ACK] Seq=4164041721 Ack=3914288462 Win=260176 Len=0
66	7.315920	64.233.169.104	192.168.1.100	TCP	1484 80 → 4335 [ACK] Seq=3914288462 Ack=4164041721 Win=8320 Len=1430 [TCP PDU reassembled in 73]

客户端收到 TCP SYN 报文的时间为 **7. 108986s**

该 SYN ACK 报文的源 IP 地址为 **64.233.169.104**、目标 IP 地址为 **192.168.1.10**

该 SYN ACK 报文的 TCP 源端口为 80，目标端口为 4335.

任务二：打开抓包文件 NAT_ISP_side 文件并回答以下问题

1.找到跟刚才客户端 7.109267s 同样目的地发送的 HTTP GET 消息。该消息何时出现在 NAT_ISP_side 跟踪文件中？承载此 HTTP GET 消息的 IP 数据报的源和目标 IP 地址以及 TCP 源和目标端口是什么?

打开 NAT_ISP_side 文件后同样进行过滤，可以发现该消息出现在 NAT_ISP_side 的时间为 6.069168.

该 IP 数据报的源 IP 地址为 71.192.34.104，目的 IP 地址为 64.233.169.104.

TCP 源端口为 4335，目标端口为 80.

如下面两图所示。

http && ip.addr == 64.233.169.104

No.	Time	Source	Destination	Protocol	Length	Info
85	6.069168	71.192.34.104	64.233.169.104	HTTP	689	GET / HTTP/1.1
90	6.117570	64.233.169.104	71.192.34.104	HTTP	814	HTTP/1.1 200 OK (text/html)
93	6.241357	71.192.34.104	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
103	6.308118	64.233.169.104	71.192.34.104	HTTP	226	HTTP/1.1 200 OK (GIF89a)
106	6.330131	71.192.34.104	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMAo4NUAILCswDjgHLCswFjgQL
121	6.407366	64.233.169.104	71.192.34.104	HTTP	648	HTTP/1.1 200 OK (text/javascript)
125	6.452270	71.192.34.104	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
131	6.496234	64.233.169.104	71.192.34.104	HTTP	870	HTTP/1.1 200 OK (text/html)
139	6.612801	71.192.34.104	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
144	6.642308	71.192.34.104	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17259,21
149	6.644609	64.233.169.104	71.192.34.104	HTTP	1359	HTTP/1.1 200 OK (PNG)
154	6.669397	71.192.34.104	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
157	6.696669	64.233.169.104	71.192.34.104	HTTP	269	HTTP/1.1 204 No Content
160	6.722203	64.233.169.104	71.192.34.104	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

```
Internet Protocol Version 4, Src: 71.192.34.104, Dst: 64.233.169.104
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 675
    Identification: 0xa2ac (41644)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 127
    Protocol: TCP (6)
    Header Checksum: 0x022f [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 71.192.34.104
    Destination Address: 64.233.169.104
    [Stream index: 7]
Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 4164040421, Ack: 3914283157,
    Source Port: 4335
    Destination Port: 80
    [Stream index: 2]
    [Stream Packet Number: 4]
  > [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 635]
    Sequence Number: 4164040421
    [Next Sequence Number: 4164041056]
    Acknowledgment Number: 3914283157
```

2.与任务一的第 3 问中找到的 HTTP GET 消息相比，此 HTTP GET 消息中的某些字段发生了变化。试分析 IP 层中的所有字段，找到发生改变的字段，说明这些字段的值由原来的什么数值改变成了现在的什么数值，并解释改变的原因。

如图所示，上图为 NAT_home_side 中的 GET 消息 IP 字段，下图为 NAT_ISP_side 中的 GET 消息 IP 字段。

发生改变的字段为:

- **源 IP 地址** source address：从原来的 192.168.1.100（私有 IP）变成了现在的 71.192.34.104（公有 IP）.

- **Time to live 字段**：从原来的 128 变成了现在的 127.
- **Header Checksum**：从原来的 0xa94a 变成了现在的 0x022f.

变化的原因：源 IP 地址发生改变的原因是 NAT 网络地址转换的结果，其将将内部私有 IP 转换为外部公有 IP，保证了内部主机在向外网发送时都使用公有 IP 进行发送。**Time to live（TTL)字段**发生改变是因为 NAT 路由器处理数据包时会消耗一个 TTL 值，因此减 1。**Header Checksum 字段**发生改变是因为，由于 IP 头部中的源地址和 TTL 字段发生了变化，校验和必须重新计算

```
Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 675
    Identification: 0xa2ac (41644)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0xa94a [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.100
    Destination Address: 64.233.169.104
    [Stream index: 5]
```

NAT_home_side 中的 GET 消息 IP 字段

```
Internet Protocol Version 4, Src: 71.192.34.104, Dst: 64.233.169.104
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 675
    Identification: 0xa2ac (41644)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 127
    Protocol: TCP (6)
    Header Checksum: 0x022f [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 71.192.34.104
    Destination Address: 64.233.169.104
    [Stream index: 7]
```

NAT_ISP_side 中的 GET 消息 IP 字段

3.在 NAT_ISP_side 跟踪文件中，从 Google 服务器收到的第一条 HTTP 200 OK 消息在什么时间？携带此 HTTP 200 OK 消息的 IP 数据报上的源 IP 和目标 IP 地址以及 TCP 源和目标端口是什么？与你在任务一的第 4 问对于回答的 NAT_home_side 的结果相比，哪些字段相同，哪些字段不同？

在 NAT_ISP_side 中，收到的第一条确认消息是在 6.117570s.

源 IP 地址：64.233.169.104，目标 IP 地址为：71.192.34.104.

TCP 源端口号为 80，目标端口号为 4335.

```
    [Time delta from previous captured frame: 0.000036000 seconds]
    [Time delta from previous displayed frame: 0.049530000 seconds]
    [Time since reference or first frame: 7.158797000 seconds]
    Frame Number: 60
    Frame Length: 814 bytes (6512 bits)
    Capture Length: 814 bytes (6512 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Ethernet II, Src: CiscoLinksys_45:1f:1b (00:22:6b:45:1f:1b), Dst: HonHaiPrecis_0d:ca:8f (00:22:68:0d:
Internet Protocol Version 4, Src: 64.233.169.104, Dst: 192.168.1.100
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x20 (DSCP: CS1, ECN: Not-ECT)
    Total Length: 800
    Identification: 0xf61e (63006)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 50
    Protocol: TCP (6)
    Header Checksum: 0xe33b [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 192.168.1.100
    [Stream index: 5]
Transmission Control Protocol, Src Port: 80, Dst Port: 4335, Seq: 3914286017, Ack: 4164041056, Len: 7
    Source Port: 80
    Destination Port: 4335
    [Stream index: 2]
    [Stream Packet Number: 8]
```

如图所示，上图为 NAT_home_side 中从 Google 服务器收到第一条消息字段，下图为 NAT_ISP_side 中从 Google 服务器收到第一条消息字段。

相同的字段：Version、Header Length、Differentiated Services Field、Total Length、Identification、Flags、Fragment Offset、Protocol、Source Address

不同的字段：

- 目的 IP 地址 Destination Address，从 192.168.1.100 变成 71.192.34.104
- Time to Live：从 50 变成 51
- Header Checksum：从 0x33b 变成 0x3a20

```
Internet Protocol Version 4, Src: 64.233.169.104, Dst: 192.168.1.100
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x20 (DSCP: CS1, ECN: Not-ECT)
    Total Length: 800
    Identification: 0xf61e (63006)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 50
    Protocol: TCP (6)
    Header Checksum: 0xe33b [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 192.168.1.100
    [Stream index: 5]
```

NAT_home_side 中从 Google 服务器收到第一条消息字段

```
Internet Protocol Version 4, Src: 64.233.169.104, Dst: 71.192.34.104
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x20 (DSCP: CS1, ECN: Not-ECT)
    Total Length: 800
    Identification: 0xf61e (63006)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 51
    Protocol: TCP (6)
    Header Checksum: 0x3a20 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 71.192.34.104
    [Stream index: 7]
```

NAT_ISP_side 中从 Google 服务器收到第一条消息字段

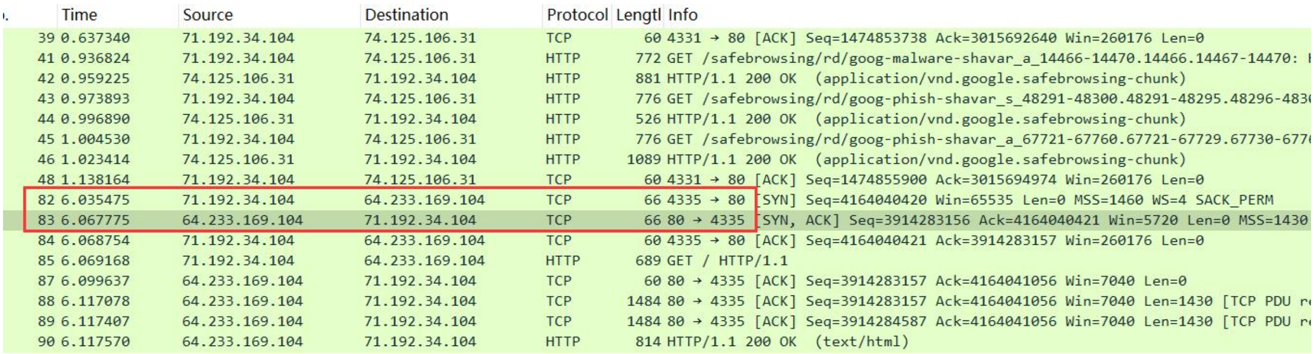
4.在 NAT_ISP_side 跟踪文件中，跟任务一的第 5 问相同的客户端到服务器 TCP SYN 报文段和服务器到客户端 TCP SYN-ACK 报文段是在什么时间出现的？这两个报文段的源 IP 和目标 IP 以及源端口和目标端口是什么？与你在任务一的第 5 问的回答相比，哪些字段相同，哪些字段与不同？

TCP SYN 报文段

- 出现的时间为：6.035475s
- 源 IP 地址为 71.192.34.104，目标 IP 地址为 64.233.169.104
- 源端口为 4335，目标端口为 80

TCP SYN-ACK 报文段

- 出现的时间为：6.067775s
- 源 IP 地址为 64.233.169.104，目标 IP 地址为 71.192.34.104
- 源端口为 80，目标端口为 4335



.	Time	Source	Destination	Protocol	Lengtl	Info
39	0.637340	71.192.34.104	74.125.106.31	TCP	60	4331 → 80 [ACK] Seq=1474853738 Ack=3015692640 Win=260176 Len=0
41	0.936824	71.192.34.104	74.125.106.31	HTTP	772	GET /safebrowsing/rd/goog-malware-shavar_a_14466-14470.14466.14467-14470:
42	0.959225	74.125.106.31	71.192.34.104	HTTP	881	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
43	0.973893	71.192.34.104	74.125.106.31	HTTP	776	GET /safebrowsing/rd/goog-phish-shavar_s_48291-48300.48291-48295.48296-483
44	0.996890	74.125.106.31	71.192.34.104	HTTP	526	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
45	1.004530	71.192.34.104	74.125.106.31	HTTP	776	GET /safebrowsing/rd/goog-phish-shavar_a_67721-67760.67721-67729.67730-677
46	1.023414	74.125.106.31	71.192.34.104	HTTP	1089	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
48	1.138164	71.192.34.104	74.125.106.31	TCP	60	4331 → 80 [ACK] Seq=1474855900 Ack=3015694974 Win=260176 Len=0
82	6.035475	71.192.34.104	64.233.169.104	TCP	66	4335 → 80 [SYN] Seq=4164040420 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
83	6.067775	64.233.169.104	71.192.34.104	TCP	66	80 → 4335 [SYN, ACK] Seq=3914283156 Ack=4164040421 Win=5720 Len=0 MSS=1430
84	6.068754	71.192.34.104	64.233.169.104	TCP	60	4335 → 80 [ACK] Seq=4164040421 Ack=3914283157 Win=260176 Len=0
85	6.069168	71.192.34.104	64.233.169.104	HTTP	689	GET / HTTP/1.1
87	6.099637	64.233.169.104	71.192.34.104	TCP	60	80 → 4335 [ACK] Seq=3914283157 Ack=4164041056 Win=7040 Len=0
88	6.117078	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=3914283157 Ack=4164041056 Win=7040 Len=1430 [TCP PDU r
89	6.117407	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=3914284587 Ack=4164041056 Win=7040 Len=1430 [TCP PDU r
90	6.117570	64.233.169.104	71.192.34.104	HTTP	814	HTTP/1.1 200 OK (text/html)

与任务一的第五问相比，在 TCP SYN 报文中：源 IP 地址、TTL 字段、Header checksum 字段不同；在 SYN-ACK 报文中，目的 IP 地址、TTL 字段、Header checksum 字段不同。其他字段均相同。

5.使用对于之前问题的回答，做出类似图 1 的 NAT 转换表

NAT 转换表

WAN 端	LAN 端
71.192.34.104, 4335	192.168.1.100, 4335

任务三：除了上面提到的 HTTP GET 消息和 HTTP 200 OK 消息以外，还与其他 Google 服务器有额外的连接，例如，在 NAT_home_side 跟踪文件中，分析时间为 1.572315s 的客户端到服务器 GET 消息，以及时间为 7.573305s 的 GET 消息。仔细研究这两个 HTTP 消息的使用，写出说明分别解释这两个消息的作用。

第一个 GET 请求如下面第一张图所示，其信息为：

/safebrowsing/rd/goog-malware-shavar_a_14466-14470.14466.14467-14470:HTTP/1.1

这条 HTTP GET 请求是 Google Safe Browsing（安全浏览）API 的一部分，用于客户端（如浏览器）从 Google 服务器获取恶意网站或恶意软件的更新数据。

10	1.269675	74.125.91.113	192.168.1.100	TCP	60	80 → 4330 [ACK] Seq=2709749796 Ack=952810709 Win=7744 Len=0
11	1.274062	74.125.91.113	192.168.1.100	HTTP	853	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-update)
12	1.474508	192.168.1.100	74.125.91.113	TCP	54	4330 → 80 [ACK] Seq=952810709 Ack=2709750595 Win=259376 Len=0
13	1.528648	74.125.91.113	192.168.1.100	TCP	853	[TCP Spurious Retransmission] 80 → 4330 [PSH, ACK] Seq=2709749796 Ack=952810709 Win=774
14	1.528673	192.168.1.100	74.125.91.113	TCP	54	[TCP Dup ACK 12#1] 4330 → 80 [ACK] Seq=952810709 Ack=2709750595 Win=259376 Len=0
15	1.529354	192.168.1.100	68.87.71.230	DNS	89	Standard query 0x1773 A safebrowsing-cache.google.com
16	1.549501	68.87.71.230	192.168.1.100	DNS	140	Standard query response 0x1773 A safebrowsing-cache.google.com CNAME safebrowsing.cache
17	1.550220	192.168.1.100	74.125.106.31	TCP	66	4331 → 80 [SYN] Seq=1474853024 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
18	1.572197	74.125.106.31	192.168.1.100	TCP	66	80 → 4331 [SYN, ACK] Seq=3015674522 Ack=1474853025 Win=5840 Len=0 MSS=1460 SACK_PERM WS
19	1.572228	192.168.1.100	74.125.106.31	TCP	54	4331 → 80 [ACK] Seq=1474853025 Ack=3015674523 Win=260176 Len=0
20	1.572315	192.168.1.100	74.125.106.31	HTTP	767	GET /safebrowsing/rd/goog-malware-shavar_s_15361-15365.15361-15365.: HTTP/1.1
21	1.601242	74.125.106.31	192.168.1.100	TCP	60	80 → 4331 [ACK] Seq=3015674523 Ack=1474853738 Win=7296 Len=0
22	1.602147	74.125.106.31	192.168.1.100	TCP	1514	80 → 4331 [ACK] Seq=3015674523 Ack=1474853738 Win=7296 Len=1460 [TCP PDU reassembled in
23	1.602464	74.125.106.31	192.168.1.100	TCP	1514	80 → 4331 [ACK] Seq=3015675983 Ack=1474853738 Win=7296 Len=1460 [TCP PDU reassembled in
24	1.602495	192.168.1.100	74.125.106.31	TCP	54	4331 → 80 [ACK] Seq=1474853738 Ack=3015677443 Win=260176 Len=0
25	1.602815	74.125.106.31	192.168.1.100	TCP	1514	80 → 4331 [ACK] Seq=3015677443 Ack=1474853738 Win=7296 Len=1460 [TCP PDU reassembled in
26	1.620968	74.125.106.31	192.168.1.100	TCP	1514	80 → 4331 [ACK] Seq=3015678903 Ack=1474853738 Win=7296 Len=1460 [TCP PDU reassembled in
27	1.621008	192.168.1.100	74.125.106.31	TCP	54	4331 → 80 [ACK] Seq=1474853738 Ack=3015680363 Win=260176 Len=0

第二条 GET 请求如下图所示。

这条 GET /generate_204 HTTP/1.1 请求是 Google 用于网络连接检测或服务状态检查的机制，常见于 Chrome 浏览器或其他 Google 服务中。/generate_204 是一个特殊的端点，由 Google 服务器提供，主要用于检测网络连通性、验证网络是否受限。

98	7.536990	64.233.169.104	192.168.1.100	TCP	1484	80 → 4335 [ACK] Seq=3914310382 Ack=4164043117 Win=11392 Len=1430 [
99	7.537332	64.233.169.104	192.168.1.100	TCP	1484	80 → 4335 [ACK] Seq=3914311812 Ack=4164043117 Win=11392 Len=1430 [T
100	7.537353	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
101	7.537381	192.168.1.100	64.233.169.104	TCP	54	4335 → 80 [ACK] Seq=4164043117 Ack=3914314058 Win=260176 Len=0
102	7.573138	74.125.91.113	192.168.1.100	TCP	66	80 → 4336 [SYN, ACK] Seq=1885299966 Ack=3887566543 Win=5720 Len=0 M
103	7.573183	192.168.1.100	74.125.91.113	TCP	54	4336 → 80 [ACK] Seq=3887566543 Ack=1885299967 Win=260176 Len=0
104	7.573305	192.168.1.100	74.125.91.113	HTTP	709	GET /generate_204 HTTP/1.1
105	7.630546	74.125.91.113	192.168.1.100	TCP	60	80 → 4336 [ACK] Seq=1885299967 Ack=3887567198 Win=7168 Len=0
106	7.631819	74.125.91.113	192.168.1.100	HTTP	179	HTTP/1.1 204 No Content
107	7.652836	192.168.1.100	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
108	7.653911	192.168.1.100	64.233.169.104	TCP	66	4337 → 80 [SYN] Seq=2326595796 Win=65535 Len=0 MSS=1460 WS=4 SACK_P
109	7.655011	192.168.1.100	64.233.169.104	TCP	66	4338 → 80 [SYN] Seq=1002769761 Win=65535 Len=0 MSS=1460 WS=4 SACK_P
110	7.682185	64.233.169.104	192.168.1.100	TCP	66	80 → 4337 [SYN, ACK] Seq=2518417000 Ack=2326595797 Win=5720 Len=0 M
111	7.682232	192.168.1.100	64.233.169.104	TCP	54	4337 → 80 [ACK] Seq=2326595797 Ack=2518417001 Win=260176 Len=0
112	7.682361	192.168.1.100	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17259,21588,21766,219